1.

2.14

$$\begin{aligned}
\int_0^\infty (1 - F_X(x))dx &= \int_0^\infty P(X > x)dx \\
&= \int_0^\infty \int_x^\infty f_X(y)dydx \\
&= \int_0^\infty \int_0^y dx f_X(y)dy \\
&= \int_0^\infty y f_X(y)dy \quad = \quad \mathrm{E}X,
\end{aligned}$$

where the last equality follows from changing the order of integration.

b.

$$\sum_{k=0}^{\infty}(1 - F_X(k)) = \sum_{k=0}^{\infty} P(X > k) = \sum_{k=0}^{\infty} \sum_{x=k+1}^{\infty} f_X(x)$$

$$= \sum_{x=1}^{\infty} \sum_{k=0}^{x-1} f_X(x) = \sum_{x=1}^{\infty} \sum_{k=1}^{x} f_X(x) = \sum_{x=1}^{\infty} x f(x)$$

$$= 0 f(x) + \sum_{x=1}^{\infty} x f(x) = \sum_{x=0}^{\infty} x f_X(x) = \mathbb{E}X$$

$\boxed{2.17}$ a) $\int_0^m 3x^2 dx = 1/2 = x^3 \Big|_0^m = m^3 \Rightarrow m = (1/2)^{1/3}$

b) ~~use the cdf~~

$$\frac{1}{\pi}\int_{-\infty}^{m} \left(\frac{1}{1+x^2}\right) dx = \frac{1}{\pi} \Big[\tan^{-1}(x)\Big]_{-\infty}^{m} = \frac{1}{\pi}\left[\tan^{-1}(m) - {-\pi/2}\right]$$

$$= \frac{1}{\pi}\tan^{-1}(m) + \frac{1}{2} = \frac{1}{2} \Rightarrow \tan^{-1}(m) = 0 \Rightarrow m = 0$$

2.20 From Example 1.5.4, if X = number of children until the first daughter, then

$$P(X = k) = (1-p)^{k-1}p,$$

where p = probability of a daughter. Thus X is a geometric random variable, and

$$\begin{aligned}
\mathrm{E}X &= \sum_{k=1}^{\infty} k(1-p)^{k-1}p = p - \sum_{k=1}^{\infty} \frac{d}{dp}(1-p)^k = -p\frac{d}{dp}\left[\sum_{k=0}^{\infty}(1-p)^k - 1\right] \\
&= -p\frac{d}{dp}\left[\frac{1}{p} - 1\right] = \frac{1}{p}.
\end{aligned}$$

Therefore, if p = $\frac{1}{2}$,the expected number of children is two.

$\boxed{2.24}$ $E(X)$, $V(X)$

(a) $E(X) = \frac{a}{a+1}$, $E(X^2) = \frac{a}{a+2}$

$V(X) = E(X^2) - (EX)^2 = \ldots =$

$$= \frac{a}{a+2} - \frac{a^2}{(a+1)^2} = \frac{a(a+1)^2}{(a+2)(a+1)^2} - \frac{a^2(a+2)}{(a+1)^2(a+2)}$$

$$= \frac{a^3 + 2a^2 + a - (a^3 + 2a^2)}{(a+1)^2(a+2)} = \frac{a}{(a+1)^2(a+2)}$$

b) $f_X(x) = \frac{1}{n}$, $x = 1, \ldots, n$

$E(X) = \sum_{x=1}^{n} \frac{1}{n} x = \frac{1}{n}\sum_{x=1}^{n} x = \frac{n+1}{2}$

$E(X^2) = \frac{1}{n}\sum_{x=1}^{n} x^2 = \frac{1}{6}(n+1)(2n+1)$

$V(X) = E(X^2) - [E(X)]^2 = \frac{1}{12}[n^2 - 1]$

c) $f_X(x) = \frac{3}{2}(x-1)^2$, $0 < x < 2$

$E(X) = 1$, $E(X^2) = 8/5$

$Var(X) = E(X^2) - (EX)^2 = 8/5 - 1 = 3/5$

5. a) the answer is $(b/\pi)^{1/2}$, which is smaller than $(b/2)^{1/2}$, the square-root of the expectation.

b) E(X^2) > (EX)^2 is larger by Jensen's Inequality – X^2 is a convex function.

c) E(log(X)) < log(EX) by Jensen's inequality – Log is concave function.

6. the answer is $e^{\mu+\sigma^2/2}$. This is larger than exponentiated expectation, e^{μ}.

7.

7. (a) $E[X] = \int_0^1 x f_X(x)\,dx = \int_0^1 2x^2\,dx = 2/3\, x^3 \Big|_0^1 = \boxed{2/3}$

(b) $E[X^2] = \int_0^1 x^2 f_X(x)\,dx = \int_0^1 2x^3\,dx = \frac{2}{4} x^4 \Big|_0^1 = \boxed{1/2}$

$\text{var}(x) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \boxed{1/18}$

(c) $E[4X+9] = 4E[X] + 9 = 4 \times 2/3 + 9 = 8/3 + 27/3 = \boxed{35/3 \text{ or } 11\, 2/3}$

(d) $E[X(1-X)] = E[X - X^2] = E[X] - E[X^2] = 2/3 - 1/2 = \boxed{1/6}$

8. $X \sim \text{Beta}(\alpha, \beta)$

$$f_X(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1} \qquad 0 \le x \le 1$$

$$E[X] = \int_0^1 x f_X(x)\,dx = \int_0^1 x \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}\,dx = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)} \underbrace{\int_0^1 \underbrace{\frac{\Gamma(\alpha+1+\beta)}{\Gamma(\alpha+1)\Gamma(\beta)} x^{(\alpha+1)-1}(1-x)^{\beta}}_{\sim \text{beta}(\alpha+1,\beta)}\,dx}_{=1}$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha+1)}{\Gamma(\alpha+\beta+1)} = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)} \cdot \frac{\alpha\Gamma(\alpha)}{(\alpha+\beta)\Gamma(\alpha+\beta)} = \boxed{\frac{\alpha}{\alpha+\beta}}$$

$$E[X^2] = \int_0^1 \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^2 x^{\alpha-1}(1-x)^{\beta-1}\,dx = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+\beta+2)} \underbrace{\int_0^1 \underbrace{\frac{\Gamma(\alpha+\beta+2)}{\Gamma(\alpha+2)\Gamma(\beta)} x^{(\alpha+2)-1}(1-x)^{\beta-1}}_{\sim \text{beta}(\alpha+2,\beta)}\,dx}_{=1}$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)} \quad \frac{\alpha(\alpha+1)\Gamma(\alpha)}{(\alpha+\beta)(\alpha+\beta+1)\Gamma(\alpha+\beta)} = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}$$

$$\text{var}(x) = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)} - \frac{\alpha^2}{(\alpha+\beta)^2} = \frac{\alpha(\alpha+1)(\alpha+\beta) - \alpha^2(\alpha+\beta+1)}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$= \frac{\alpha^3 + \alpha^2\beta + \alpha^2 + \alpha\beta - \alpha^3 - \alpha^2\beta - \alpha^2}{(\alpha+\beta)^2(\alpha+\beta+1)} = \boxed{\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}}$$

2.30 a) $E[e^{tx}] = \frac{1}{c}\int_0^c e^{tx}\,dx = \frac{1}{ct} e^{tx}\Big|_0^c$

$$= \begin{cases} \frac{e^{c}-1}{tc}, & t \neq 0 \\ 1 & t = 0 \end{cases}$$

b) $E(e^{tx}) = \frac{2}{c^2}\int_0^c x e^{tx}\,dx = \frac{2}{c^2}\left[\frac{e^{tx}(tx-1)}{t^2}\right]\Big|_0^c$

$= \frac{2}{c^2}\left[\frac{e^{tc}(tc-1)}{t^2} - \frac{-1}{t^2}\right] = \frac{2}{c^2t^2}\left[1 + e^{tc}(tc-1)\right]$

$t: e^{tc}(tc-1) > -1$

c) $\int_{-\infty}^{\infty} \frac{1}{2\beta} e^{-|x-\alpha|/\beta} e^{tx}\,dx$ Let $Y = \frac{X-\alpha}{\beta}$

$M_Y(t) = \frac{1}{2}\int_{-\infty}^{\infty} e^{ty} e^{-|Y|}\,dy = \frac{1}{2}\int_{-\infty}^{0} e^{ty+y}\,dy + \frac{1}{2}\int_0^{\infty} e^{ty-y} = \frac{1}{2}\left[\frac{1}{t+1} e^{y(t+1)}\Big|_{-\infty}^{0} + \right.$

$\left.\frac{1}{t-1}\exp(y(t-1))\Big|_0^{\infty}\right] = \frac{1}{2}\left[\frac{1}{t+1}(1-0) + \frac{1}{t-1}\cdot[0-1]\right]$ if $t<1$

$= \frac{1}{2}\left[\frac{(t-1) - (t+1)}{t^2-1}\right] = \frac{-1}{t^2-1}$

$= \frac{1}{1-t^2}$. $X = \beta Y + \alpha$, So

$M_x(t) = e^{\alpha t} M_Y(\beta t) = \boxed{\frac{e^{\alpha t}}{1-\beta^2 t^2}}$, $|t| < 1/\beta$

d) $E(e^{tx}) = \sum_{x=0}^{\infty} e^{tx}\binom{r+x-1}{x} p^r (1-p)^x$

$= p^r \sum_{x=0}^{\infty}\binom{r+x-1}{x}\left[e^{tx}(1-p)\right]^x$.

$= \frac{p^r}{(1-(1-p)e^t)^r} \sum_{x=0}^{\infty}\binom{r+x-1}{x}\left[e^{t}(1-p)\right]^x\left[1-e^t(1-p)\right]^r$ 1

$= \left(\frac{p}{1-(1-p)e^t}\right)^r$ $t < -\log(1-p)$.

2.33

(a) $$\sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda}\sum_{x=0}^{\infty}\frac{(e^t\lambda)^x}{x!} = e^{-\lambda}e^{\lambda e^t} = e^{\lambda(e^t-1)}$$

$$EX = M_x'(t)\Big|_{t=0} = e^{\lambda(e^t-1)}\cdot\lambda e^t\Big|_{t=0} = e^{\lambda(1-1)}\lambda e^0 = \lambda$$

$$EX^2 = M_x''(t) = \lambda e^t e^{\lambda(e^t-1)} + \lambda^2 e^{2t} e^{\lambda(e^t-1)}\Big|_{t=0} = \lambda + \lambda^2$$

$$\Rightarrow Var(X) = E(X^2) - (EX)^2 = \lambda + \lambda^2 - \lambda^2 = \lambda$$

(b) $$\sum_{x=0}^{\infty} e^{tx} p(1-p)^x = p\sum_{x=0}^{\infty}\left(e^t(1-p)\right)^x = p\left(\frac{1}{1-e^t(1-p)}\right)$$

$$EX = M_x'(t)\Big|_{t=0} = -\frac{p}{(1-e^t(1-p))^2}\cdot\left(-(1-p)e^t\right)\Big|_{t=0} = \frac{p(1-p)e^t}{(1-e^t(1-p))^2}\Big|_{t=0} = \frac{p(1-p)}{p^2} = \frac{(1-p)}{p}$$

$$EX^2 = M_x''(t)\Big|_{t=0} = \frac{p(1-p)e^t}{(1-e^t(1-p))^2} - 2\frac{p(1-p)e^t}{(1-e^t(1-p))^3}\cdot\left(-(1-p)e^t\right)\Big|_{t=0}$$

$$= p(1-p)\left[\frac{e^t}{(1-e^t(1-p))^2} + 2\frac{(1-p)e^{2t}}{(1-e^t(1-p))^3}\right]\Big|_{t=0} = p(1-p)\left[\frac{1}{p^2} + 2\frac{(1-p)}{p^3}\right]$$

$$= \frac{(1-p)}{p} + 2\frac{(1-p)^2}{p^2} = \frac{p(1-p) + 2(1-p)^2}{p^2}$$

$$V(X) = E(X^2) - (EX)^2 = \frac{p(1-p) + 2(1-p)^2 - (1-p)^2}{p^2}$$

$$= \frac{p(1-p) + (1-p)^2}{p^2} = \frac{p - \cancel{p^2} + 1 + \cancel{p^2} - 2p}{p^2} = \frac{1-p}{p^2}$$

c) $$M_x(t) = E(e^{tx}) = \int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{1}{2\sigma^2}\left(x^2 - 2\mu x + \mu^2 - 2\sigma^2 tx\right)\right\}$$ $\Rightarrow$ complete square

$$e^{\frac{1}{2\sigma^2}(2\mu\sigma^2 t + (\sigma^2 t)^2)}\frac{1}{\sqrt{2\pi\sigma^2}}\int_{-\infty}^{\infty}\exp\left\{-\frac{1}{2\sigma^2}\left(x - (\mu + \sigma^2 t)\right)^2\right\}dx = e^{\frac{1}{2\sigma^2}(2\mu\sigma^2 t + \sigma^4 t^2)}$$

$$= e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

$$EX = M'_X(t)\Big|_{t=0} = (\mu+\sigma^2 t)e^{\mu t+\sigma^2t^2/2}\Big|_{t=0} = \mu$$

$$EX^2 = M''_X(t)\Big|_{t=0} = (\mu+\sigma^2 t)^2 e^{\mu t+\sigma^2t^2/2} + \sigma^2 e^{\mu t+\sigma^2t^2/2}\Big|_{t=0}$$

$$= \mu^2+\sigma^2$$

$$Ver(X) = \mu^2+\sigma^2-\mu^2 = \sigma^2$$

$\boxed{2.38}$ (a) $M_X(t) = \sum_{x=0}^{\infty} e^{tx}\binom{r+x-1}{x}p^r(1-p)^x$

$$= \frac{p^r}{(1-e^t(1-p))^r}\sum_{x=0}^{\infty}\binom{r+x-1}{x}(1-e^t(1-p))^r(e^t(1-p))^x$$

$$= p^r \Big/ (1-e^t(1-p))^r$$

(b) $Y = 2pX \Rightarrow$ ~~$X = Y/2p$~~

$$M_Y(t) = M_X(2pt) = \frac{p^r}{(1-e^{2pt}(1-p))^r} = \left(\frac{p}{1-e^{2pt}(1-p)}\right)^r$$

$$\lim_{p\downarrow 0} M_Y(t) = \lim_{p\downarrow 0}\left(\frac{p}{1-e^{2pt}(1-p)}\right)^r$$

$$= \lim_{p\downarrow 0}\left(\frac{1}{1-2t(1-p)e^{2pt}}\right)^r \quad \text{by L'hospital's rule}$$

$$= \left(\frac{1}{1-2te^0}\right)^r = \left(\frac{1}{1-2t}\right)^r \quad \text{for } t<1/2$$

~~4.~~ 12.

a) $e^{\mu t+\sigma^2t^2/2}$

b) $e^{\mu t/n+\sigma^2t^2/2n}$

~~5.~~ 13. a. The Answer is p

b.

$$M_x(t) = E(e^{tx}) = (1-p)e^{t\cdot 0} + \int^{\infty} p e^{-x} e^{tx}\,dx$$

$$= (1-p) + p\int_0^{\infty} e^{-x(1-t)}\,dx$$

$$= (1-p) + p\int_0^{\infty} \frac{(1-t)}{(1-t)} e^{-x(1-t)}\,dx$$

$$= (1-p) + \frac{p}{(1-t)} \quad \text{for } t < 1$$